USB POWER 45W S8.5 / SINGLE BOARD, +5V SERIES PROTECTION

S8.5: +5V fast series disconnect (S8.4, FLT off the latch); +/−12V directly from converters; ceramic banks; two parallel outputs; VPD-powered OV guard; one-shot arming latch; MAIN LED only.

USB PD, main eFuse and VIN damper

RUN
VPD
VIN
MAIN_OK

CH224A: 20 V request
TPS16630: 4.186 A common input switch
VIN damper 1R + 2 x 10u

File: 01_usb_input.kicad_sch

+12 V converter

MAIN_OK
VIN
+12V

TPS62933F
+12 V delivered rail

File: 02_positive_12v.kicad_sch

+5 V converter

MAIN_OK
VIN
P5_RAW

TPS62933F
P5_RAW before disconnect

File: 03_positive_5v.kicad_sch

−12 V inverting converter

+12V
VIN
−12V

LMR38020F inverting buck-boost
−12 V delivered rail

File: 04_negative_converter.kicad_sch

Two output headers — monitors, overvoltage guard and MAIN LED

+12V
−12V
MAIN_LED_OK
P5_RAW
RUN
V3V3
VPD
P12_OK
P5_OK
N12_OK
V20_OK
OV_N

Two Eurorack outputs J2 + J3
Monitors, OV guard, +5V eFuse
MAIN LED (only status LED)

File: 06_outputs_status.kicad_sch

FAULT POLICY
One-shot qualification from the first converter enable (STARTED latch): after
0.31–0.88 s any persistent bad status -> all OFF until RESET. Severe OV via U20 acts even before ARMED.
U23 independently cuts +5V; OVLO alone is auto-recovering. U23 FLT is not latched.
MAIN indicates enabled/status, not precision regulation.

OUTPUTS: +12 / +5 / −12 on J2 and J3 in parallel
Published conservative rating (S8.4, prototype): 2.5 / 2.0 / 0.9 A.
<=45 W total; above 2 A per rail use both headers. U23 ILIM 3.03 A nominal is a threshold, not a rating.

Always-on control — fault latch, overvoltage preset and RESET

MAIN_OK
N12_OK
OV_N
P12_OK
P5_OK
V20_OK
VPD
RUN
V3V3
MAIN_LED_OK

60 V always-on LDO / one-shot STARTED + FAULT latches
Manual RESET; OV_N presets FAULT

File: 05_fault_reset.kicad_sch

One-shot arming latch (STARTED): U18/U20 on VPD; see docs/s8-single-board.md
S8.5 single board: +5V eFuse as S8.4; +/−12V unswitched; MAIN LED only

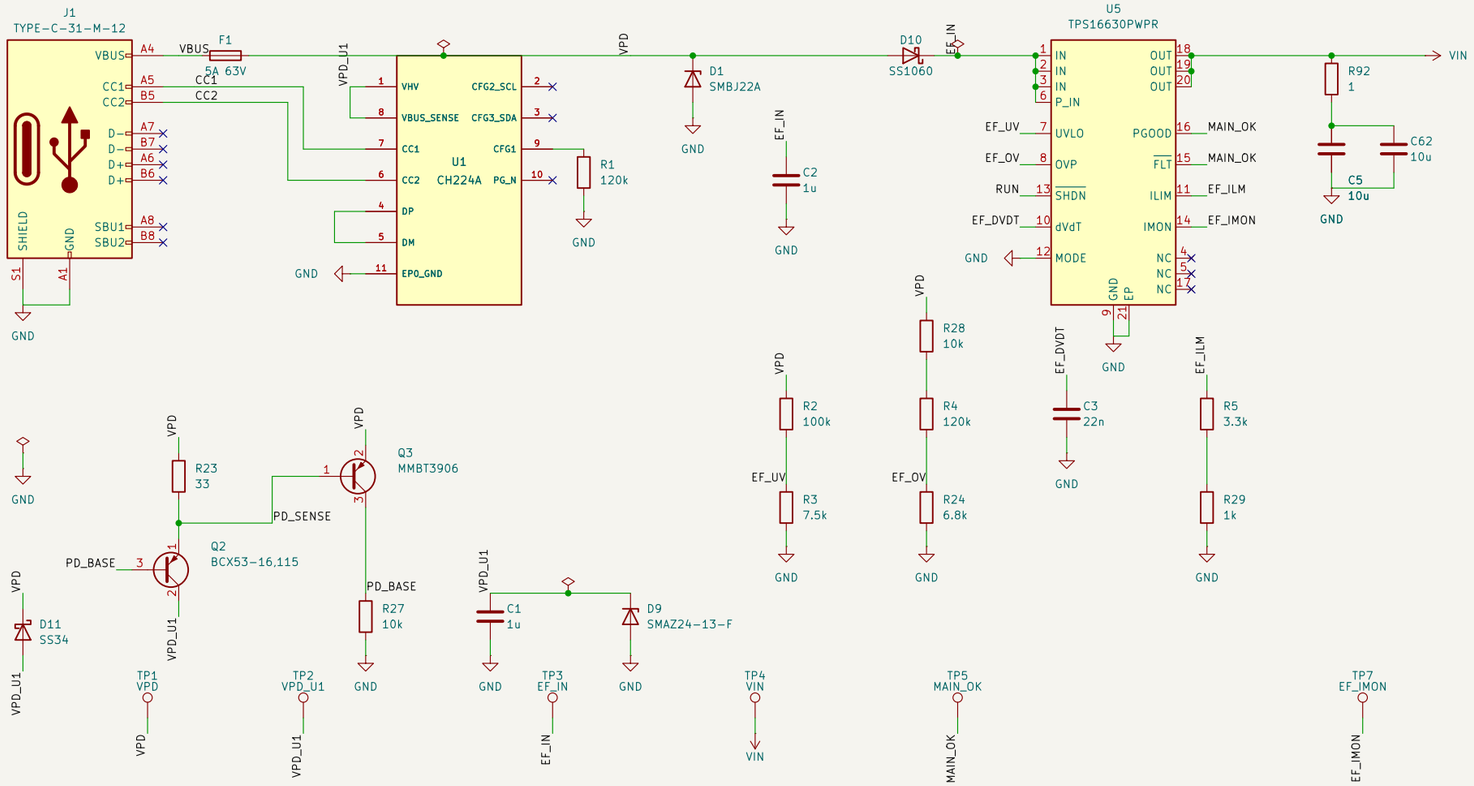
Big Modular

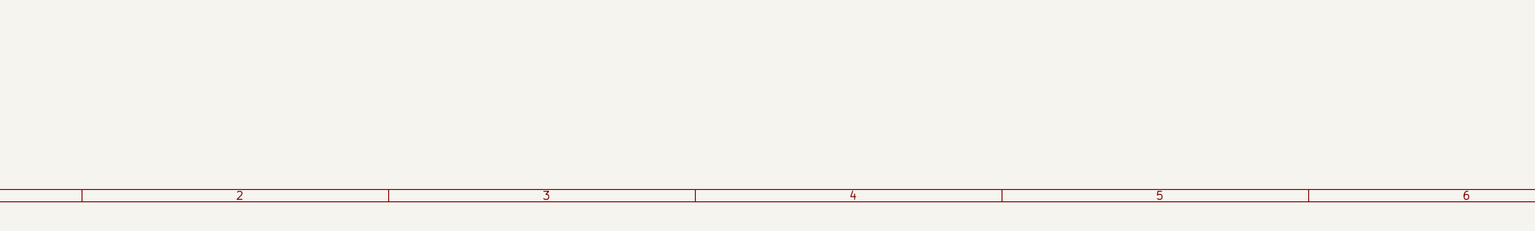
Sheet: /
File: usb-power-45w-single.kicad_sch

Title: USB power S8 single board — wired power flow overview

Size: User	Date:	**Rev: S8.5**
KiCad E.D.A. 10.0.6		Id: 1/7

USB PD, MAIN EFUSE AND VIN DAMPER





20 V PD request: U5 TPS16630 (60 V, 4.186 A nominal) is the common input switch: RUN low turns every rail off.
MAIN_OK = FLT + PGOOD at 3.3 V. R92 1R + C5/C62 2 x 10u damp the all-ceramic VIN bus (replaces the 47u can).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK
RUN
VIN
VPD

S8.5: S8.4 +5V eFuse kept, one-shot arming latch — docs/s8-single-board.md

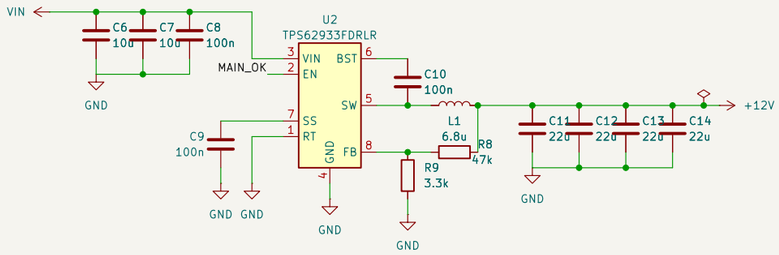
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Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

Title: USB PD, main eFuse and VIN damper

Size: User	Date:	**Rev: S8.5**
KiCad E.D.A. 10.0.6		Id: 2/7

+12 V CONVERTER





TPS62933F +12 V, unchanged power stage; its output IS the delivered +12V (no rail eFuse).
Soft start C9 100n = ~14.5 ms into rack capacitance. Own protection: 4.2–5.8 A peak limit, hiccup, OVP 112–118 %, TSD.
Rail ceramics: C11–C14 here + C34 (J2) and C59 (J3) on page 06; the 100u can is gone.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇ +12V

MAIN_OK ◇ MAIN_OK

VIN ◇ VIN

S8.5: S8.4 +5V eFuse kept, one-shot arming latch — docs/s8-single-board.md

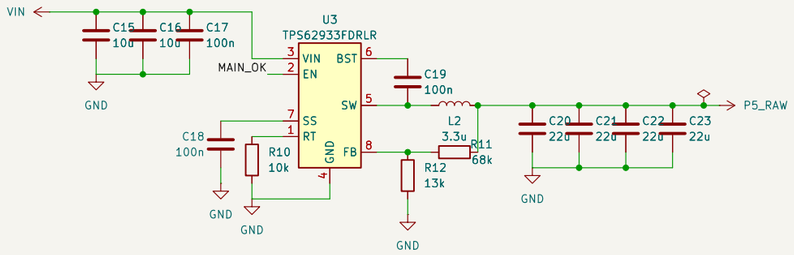
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Sheet: /+12 V converter/
File: 02_positive_12v.kicad_sch

Title: +12 V converter

Size: User	Date:	**Rev: S8.5**
KiCad E.D.A. 10.0.6		Id: 3/7

+5 V CONVERTER





TPS62933F +5 V: P5_RAW before U23. Feedback R11/R12 (0.1 %, S8.4) and C20–C23 remain upstream; never sense after U23.
C69–C72/C75/C76 extend the upstream bank to 10 x 22u. Delivered +5V has C35/C60/C73/C74 after U23.
C18 100n converter soft start; U23 has its own dVdt ramp. Converter/load stability is not bench qualified.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇——— MAIN_OK

P5_RAW ◇——— P5_RAW

VIN ◇——— VIN

S8.5: S8.4 +5V eFuse kept, one-shot arming latch — docs/s8-single-board.md

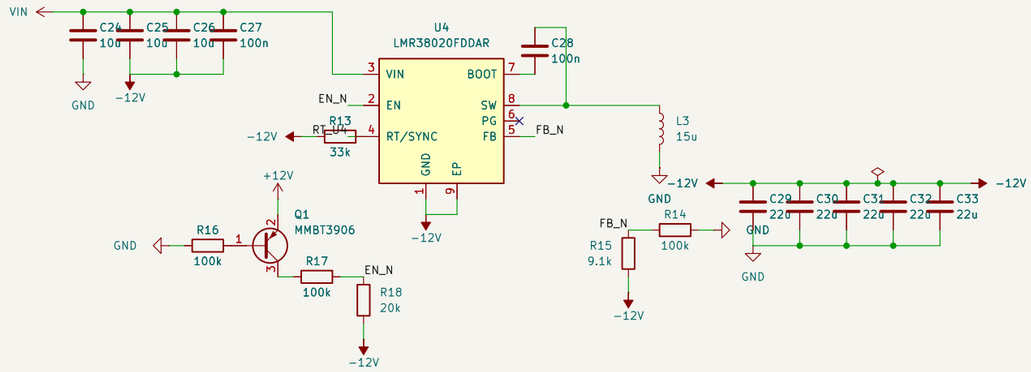
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Sheet: /+5 V converter/
File: 03_positive_5v.kicad_sch

Title: +5 V converter

Size: User	Date:	**Rev: S8.5**
KiCad E.D.A. 10.0.6		Id: 4/7

−12 V INVERTING CONVERTER





LMR38020F inverting buck−boost; IC ground / exposed pad = the delivered −12V (no series switch).
4 ms soft start, 18 ms OCP blanking at start−up. Q1 follows +12V. C29−C33 here + C36 (J2) and C61 (J3).
S8.4: hiccup needs |Vout| < 40 % after the blanking. Rack allowance 0.85 mF (TI 1000 uF COUT guidance; hiccup model
1.08 mF at 15.2 V converter input, 1 A) covers a typical 0.44 mF rack. Bench start−up test still required.

INTER−SHEET INTERFACE (see wired power−flow overview)

+12V ◇ +12V

−12V ◇ −12V

VIN ◇ VIN

S8.5: S8.4 +5V eFuse kept, one−shot arming latch − docs/s8−single−board.md

Big Modular

Sheet: /−12 V inverting converter/
File: 04_negative_converter.kicad_sch

Title: −12 V inverting converter

Size: User	Date:	**Rev: S8.5**
KiCad E.D.A. 10.0.6		Id: 5/7

ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET

U9
TPS7A1633DGNR
C46 1u
C47 10u
R63 10k
MAIN_OK

R64 10k
RUN

U10
TPS3808G01DBVR
R91 15k
R65 47k
R66 10k
POR_SEN
R67 10k
R68 10k
RESET_BTN_N
C48 100n

U11
TPS3808G01DBVR
R69 33k
TIMER_GO
R70 10k
START_SENSE
C49 100n
START_CT
R71 10k
ARMED
C50 100n

SW1
TL3340AF160QG
RESET_BTN_N

U12
SN74HC74DR
C52 100n

U13
SN74HC08DR
C53 100n

U14
SN74HC20DR
C54 100n

R72 10k
ANY_BAD
BAD_RC
C51 100n

U15
SN74LVC1G17DBVR
C55 100n

TP9 RUN
TP10 ARMED
TP11 FAULT_LATCH
TP12 P12_OK
TP13 P5_OK
TP14 N12_OK
TP15 V20_OK
TP16 V3V3
TP6 OV_N
TP19 STARTED

RESET clears STARTED and FAULT; RUN is forced low while reset is asserted.
S8.5 one-shot arming: START_N = NAND(MAIN_LED_OK, RUN) (U14 gate 2) presets STARTED (U12 A) at the first converter enable; only RESET clears it. TIMER_GO = RUN AND STARTED -> U11 -> ARMED 0.31–0.88 s later, held until RESET/FAULT: no fault can restart or extend the window. Then any persistent bad status latches every rail OFF. OV_N presets FAULT at any time.
V20_OK (R83 120k, 9.06–10.96 V) only validates U5 FLT/PGOOD (MAIN_OK): the input qualification is the U5 UVLO.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_LED_OK
MAIN_OK
N12_OK
OV_N
P12_OK
P5_OK
RUN
V20_OK
V3V3
VPD

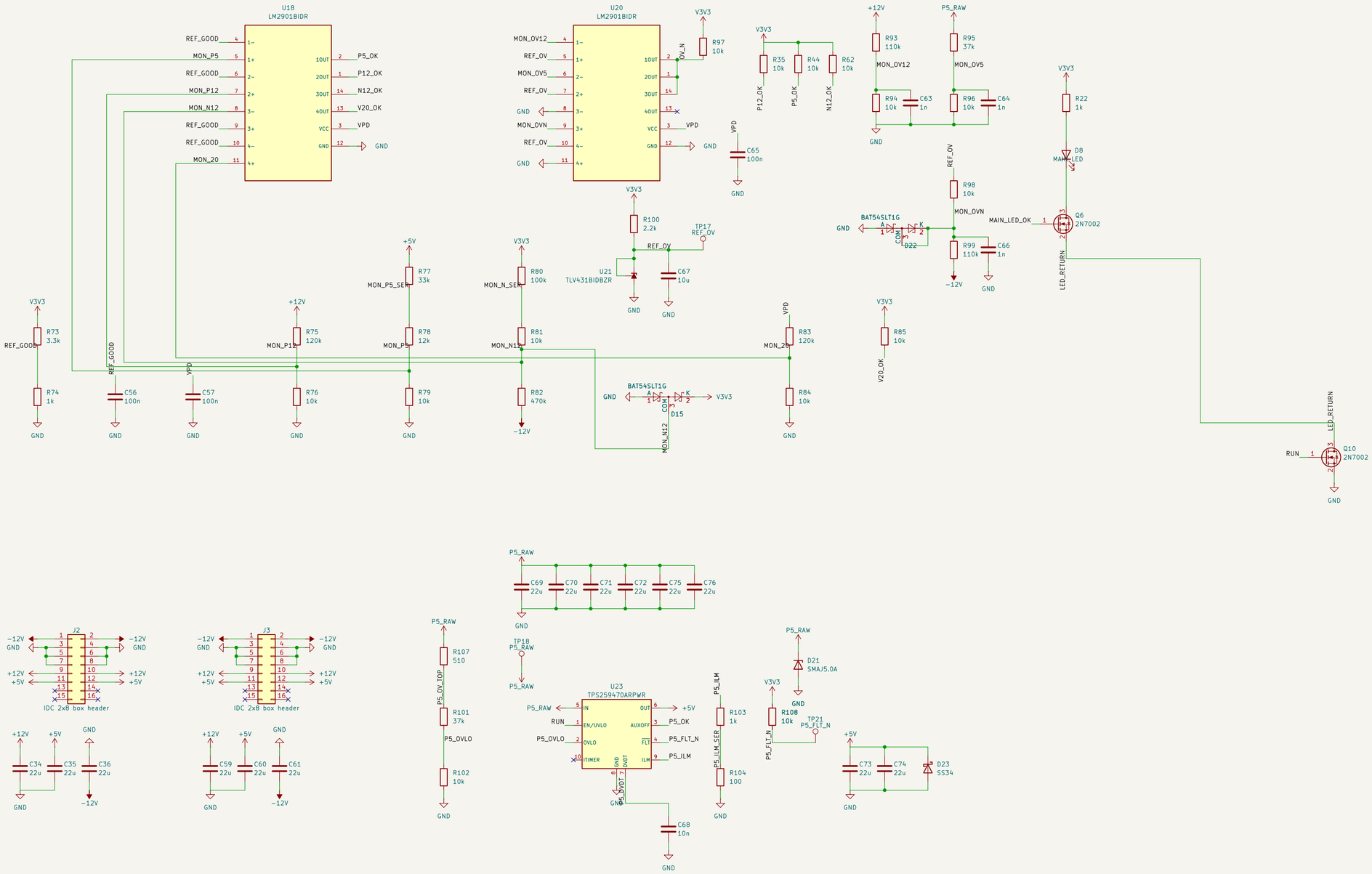
S8.5: S8.4 +5V eFuse kept, one-shot arming latch — docs/s8-single-board.md
Big Modular
Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 05_fault_reset.kicad_sch

Title: Always-on control — fault latch, overvoltage preset and RESET

Size: User	Date:	**Rev: S8.5**
KiCad E.D.A. 10.0.6		Id: 6/7

TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

J2 + J3: two parallel 16-pin outputs, pin 1 = −12V. C34–C36 at J2, C59–C61 at J3 (22u each, replace the 100u cans).
D8 MAIN is the only status LED (rail LEDs removed): MAIN_LED_OK (U5 on, FLT/PGOOD valid) AND RUN. Any bad status after ARMED or a U20 guard trip latches every rail off. Short OVLO–only pulses may recover. P12/P5/N12_OK: logic + test pads.
U20 overvoltage guard (S8.1): +12V > 14.88 V, +5V > 5.83 V or −12V < −13.64 V nominal (corners 14.48–15.27 / 5.68–5.98 / 13.27–14.02 V) pulls OV_N → FAULT → U5 off. Own reference REF_OV = U21 TLV431B (R100, C67, TP17); U18/U20 LM2901B: R93–R96, R98, R99 are 0.1 % / 25 ppm. The −12V channel compares MON_OVN with GND (no V3V3 term): D22 holds MON_OVN near −0.24 V (25 C only, <=0.15 mA) if V3V3 is lost while the rack holds −12V. D21 SMAJ5.0A is a transient clamp, not an OV limit.
S8.3: U18/U20 supplies are always-on VPD: outputs remain 3.3 V logic, U20 +5V guard monitors P5_RAW.
U23 TPS259470A independently disconnects +5V at ~5.70 V (1.2 us typical, no maximum specified). D21 moved to P5_RAW.
S8.4: P5_OK = AUXOFF + U18 only: FLT (reverse current/overload/OT) goes to R108/TP21, never to the latch.
Severe raw OV is unmasked through U20. No guaranteed 6 V header peak or prototype qualification is claimed.



INTER-SHEET INTERFACE (see wired power-flow overview)

+12V — +12V
−12V — −12V
MAIN_LED_OK — MAIN_LED_OK
N12_OK — N12_OK
OV_N — OV_N
P12_OK — P12_OK
P5_OK — P5_OK
P5_RAW — P5_RAW
RUN — RUN
V20_OK — V20_OK
V3V3 — V3V3
VPD — VPD

S8.5: S8.4 +5V eFuse kept, one-shot arming latch — docs/s8-single-board.md
Big Modular
Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 06_outputs_status.kicad_sch

Title: Two output headers — monitors, overvoltage guard and MAIN LED

Size: User	Date:	**Rev: S8.5**
KiCad E.D.A. 10.0.6		Id: 7/7